

# Non-Parametric Statistics

## Week 04 Lecture Notes

### Lectures 5–6

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#### L-5 Multiple Samples

##### Parametric Case

We have  $m$  samples drawn from  $m$  populations:

$$\begin{aligned} X_{11}, \dots, X_{1n_1} &\stackrel{\text{i.i.d.}}{\sim} F_1(\mu_1) \\ X_{21}, \dots, X_{2n_2} &\stackrel{\text{i.i.d.}}{\sim} F_2(\mu_2) \\ &\vdots \\ X_{m1}, \dots, X_{mn_m} &\stackrel{\text{i.i.d.}}{\sim} F_m(\mu_m) \end{aligned}$$

*m samples from m populations*

Null hypothesis:

$$H_0 : \mu_1 = \dots = \mu_m$$

ANOVA needs to assume  $F_i$  are normal.

##### Non-Parametric Case

General  $F$ , testing for equality of the sample **medians**.

Put all  $X_{ij}$ 's together and **rank**.

For sample  $j$  ( $j = 1, \dots, m$ ), take the corresponding ranks (elements coming from the  $j$ -th sample).

Sum those ranks  $R_j$ ,  $j = 1, \dots, m$ .

Under  $H_0$ ,

$$\frac{R_j}{n_j} \longrightarrow \text{these should be close.}$$

There should be **no difference** (or very little) between the average ranks of different groups.

**Test statistic** is the variance of  $R_j/n_j$ :

$$\sum_j n_j \left( \frac{R_j}{n_j} - \bar{R} \right)^2$$

where  $n = \sum_{j=1}^m n_j$ , and

$$\bar{R} = \text{avg rank for all samples together} = \frac{n(n+1)}{2n} = \frac{n+1}{2}.$$

### Kruskal–Wallis Statistic & Test

$$\begin{aligned} H &= \frac{12}{n(n+1)} \left[ \sum_{j=1}^m \frac{R_j^2}{n_j} - \frac{(n+1)^2 n}{4} \right] \\ &= \frac{12}{n(n+1)} \sum_{j=1}^m \frac{R_j^2}{n_j} - 3(n+1). \end{aligned} \quad (1)$$

$H \approx \chi_{m-1}^2$  (approximately);  $n_j$  should be large for each  $j$ .

The *exact* distribution of the test statistic can be derived by **permutation**.

$$n = n_1 + n_2 + \cdots + n_m$$

$$H_1, \dots, H_K$$

Number of permutations:

$$\frac{n!}{n_1! n_2! \cdots n_m!} \quad (\text{many permutations})$$

Example with  $\binom{14}{5} \binom{9}{7}$ :

$$\binom{14}{5} \binom{9}{7} = \frac{14!}{5! 7! 2!}$$

1. Calculate  $H$  for each permutation.
2. Find p-value from the position of the observed  $H$  in the histogram.

### Kruskal–Wallis Test

**Example** (Hollander & Wolfe, 1999) — *Mucociliary Efficiency*.

Foam rate of diet in  $m = 3$  groups:

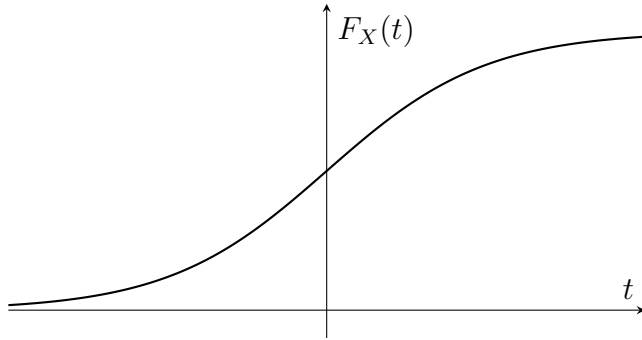
- Normal
- Obstructive airway
- Asbestosis

$n_j$ : 5, 4, 5;  $n = 14$ .

**HW:** Write R code for this problem.

## Inference for CDF

The family  $\{F_x(b), -\infty < b < \infty\}$  is the parameter — an **infinite-dimensional** parameter.



$X \rightarrow$  random variable;  $F_x(t) = P(X \leq t)$ ;  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F_x$ .

For  $t = 5$ :  $P(X \leq 5) = \theta$  is estimated by  $\frac{1}{n} \#\{x_i \leq 5\}$ , giving estimate  $\hat{\theta}$ .

### **Empirical CDF**

$$\hat{F}_n(t) = \frac{1}{n} \#\{x_i \leq t\}$$

For  $\theta = F_x(t)$ , a point estimate is  $\hat{\theta} = \hat{F}_n(t)$ .

### **Distribution of $\hat{\theta}$ :**

$$Y(t) = \#\{x_i \leq t\} = \sum_{i=1}^n \mathbf{1}(x_i \leq t) \sim \text{Bin}(n, \theta), \quad P(x_i \leq t) = F_x(t) = \theta.$$

If  $n$  is large enough, we can use the CLT for the binomial.

### **CI for $\theta$ :**

$$\left( \hat{\theta} \pm z_{\alpha/2} \sqrt{\frac{\hat{\theta}(1 - \hat{\theta})}{n}} \right)$$

$$\hat{\theta} = \frac{Y(t)}{n} = \frac{1}{n} \#\{x_i \leq t\}.$$

*Note:* The estimate of  $F$  is a step function (discontinuous), even when  $F$  is continuous.

**Order Statistics and the Empirical CDF**

Order statistics:  $X_{(1)} < \dots < X_{(n)}$ .

$$\hat{F}_n(t) = \begin{cases} 0 & t < X_{(1)} \\ \frac{1}{n} & X_{(1)} \leq t < X_{(2)} \\ \frac{2}{n} & X_{(2)} \leq t < X_{(3)} \\ \vdots & \\ \frac{n-1}{n} & X_{(n-1)} \leq t < X_{(n)} \\ 1 & X_{(n)} \leq t \end{cases}$$

Changes at  $n + 1$  points. No need to compute for every  $t$ .

CDF, PDF; quantile function  $F^{-1}(x)$ .

Quantile function:  $\{F^{-1}(x)\}$ ,  $0 \leq x \leq 1$  a parameter.

**Testing / Estimation of Median**

When  $x = \frac{1}{2}$ :  $\theta = \text{median}$ ;  $\hat{\theta} = \text{sample median}$ .

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_a : \theta \neq \theta_0.$$

$$\frac{\hat{\theta} - \theta_0}{\sqrt{\frac{\theta_0(1 - \theta_0)}{n}}} \sim N(0, 1).$$

Also derive an **exact test** for the median.

When  $x = \frac{3}{4}$ , the third percentile =  $\hat{\theta}$ :  $\hat{\theta} = \text{sample 3}^{\text{rd}}$  quartile.

$$\hat{\theta} = \min\{x : 0.75n \text{ of the } x_i\text{'s are } \leq x\}.$$

CI and test for quantiles.

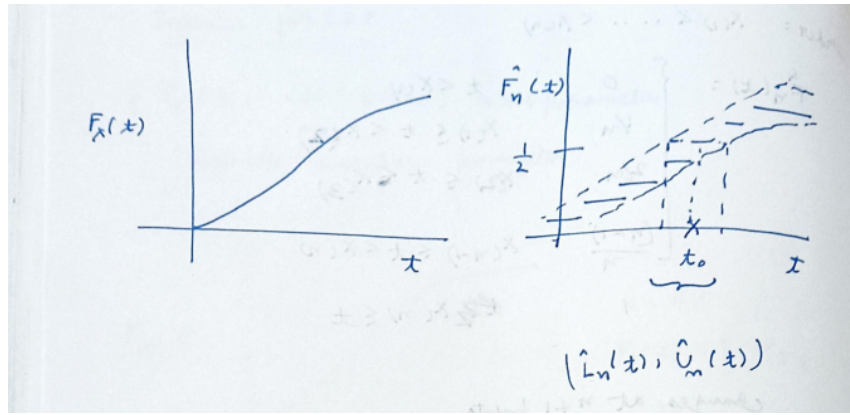


Figure 1: CDF and its Confidence intervals

## L-6 Testing for Median

Is the population median 0.2?

Sample  $X_1, \dots, X_n$ .

$$H_0 : P(X > 0.2) = \frac{1}{2} = p_0 \quad H_a : p \neq p_0.$$

$$\hat{p} = \frac{1}{n} \# \{x_i > 0.2\}.$$

Under  $H_0$ :  $n\hat{p} \sim \text{Bin}(n, \frac{1}{2})$ .

Either **binomial test** or **normal approximation** if  $n$  is sufficiently large.

Is the Third Quartile 0.3?

$$H_0 : P(X > 0.3) = 0.25 = p_0 \quad \text{or equivalently} \quad H_0 : P(X < 0.3) = 0.75.$$

$$H_a : p \neq p_0, \quad p = P(X > 0.3).$$

$$\hat{p} = \frac{1}{n} \# \{x_i > 0.3\}.$$

Under  $H_0$ :  $n\hat{p} \sim \text{Bin}(n, 0.25)$ .

p-value or critical region; use exact binomial or normal approximation.

**Example.** Given 176 observations out of 250 are less than  $a = 1.5$ . Test if 1.5 is the third quartile.

$$p = P(X < 1.5), \quad H_0 : p = 0.75, \quad H_a : p \neq 0.75, \quad \alpha = 5\%.$$

$$\hat{p} = \frac{176}{250}.$$

$n\hat{p} \sim \text{Bin}(n, p_0) = \text{Bin}(250, 0.75).$

$$Z = \frac{176 - 250 \times 0.75}{\sqrt{250 \times 0.75 \times 0.25}} = -1.68.$$

p-value from  $(-1.68) = 0.0929.$

We do **not** reject at the 5%  $\alpha$ -level.

$$2 \times p_{\text{binom}}(176, 250, 0.75) = 0.1119 \quad (\text{exact}).$$

## Estimating the CDF

Estimate CDF:  $F_x(t) = P(X \leq t) = \theta.$

$$\hat{\theta} = \frac{1}{n} \# \{x_i \leq t\}.$$

$n\hat{\theta} \sim \text{Bin}(n, \theta).$

**CI for  $\theta$ :**

$$\left( \hat{\theta} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} \right).$$
$$\hat{F}_n(t) = \begin{cases} 0 & t < X_{(1)} \\ 1/n & X_{(1)} \leq t < X_{(2)} \\ \vdots & \\ k/n & X_{(k)} \leq t < X_{(k+1)} \\ 1 & t > X_{(n)} \end{cases}$$

Not continuous; each step height is  $\frac{1}{n}.$

## Confidence Intervals for CDF and Quantile Functions

= Testing **goodness of fit** of a distribution.

$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F; \quad H_0 : F = F_0.$

**Case 1 —  $F_0$  discrete ( $\chi^2$  test):**

$$\chi^2 = \sum \frac{(\text{Obs} - \text{Exp})^2}{\text{Exp}}$$

**Binning:** histogram / frequency table. For continuous distributions, there is *no* obvious / objective choice of binning.

Each bin has same probability under  $H_0$ .

## Kolmogorov–Smirnov Statistic

Alternatively, look at the statistic

$$D = \max_{-\infty < t < \infty} |\hat{F}_n(t) - F_0(t)| = \max_{t = X_{(1)}, \dots, X_{(n)}} \{|\hat{F}_n(t) - F_0(t)|\}$$

(Kolmogorov–Smirnov statistic).

*Note:* the maximum occurs at one of the data points; max step size =  $\frac{1}{n}$ .

1. Evaluation is taken at **finitely many** points.
2.  $D$  is **distribution-free**.

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F_0$$

Continuous distribution  $F_0$  increasing  $F(X) \sim \text{Unif}(0, 1)$ .

Let  $S = \hat{F}_n(t) - F_0(t)$ :

$$\begin{aligned} S &= \frac{1}{n} \#\{x_i \leq t\} - F_0(t) \\ &= \#(i : F_0(x_i) \leq F_0(t)) \cdot \frac{1}{n} - F_0(t) \\ &= \#(i : U_i \leq F_0(t)) \cdot \frac{1}{n} - F_0(t) \quad U_1, \dots, U_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1) \\ &= \#(i : U_i \leq u) \cdot \frac{1}{n} - u \quad [F_0(t) = u] \\ &= \tilde{F}_n(u) - u \end{aligned}$$

where  $\tilde{F}_n$  is the ECDF of  $U_1, \dots, U_n$ .

The RHS does *not* depend on  $F_0$ , so it is **distribution-free**.

Enough to derive the distribution of  $D$  when  $F_0 = \text{Unif}(0, 1)$ .